

Audit evidence — data / code / results provenance

Public repo: <https://github.com/Coucou2016/4DVarNets-sea-current>
Purpose: Demonstrate that reported numbers are computed in *this* repository, quarantine legacy protocols, and list what was **not** run. This document is *not* a manuscript section.
Audit date: 2026-09-15

1. Scientific honesty rules (enforced)

1. Use **only** metrics JSON under `results/` (including `results/legacy_pre_review2/` when labelled as quarantine).
2. **Never** copy Fablet/JAMES (or any external) table numbers as “our results.”
3. Label protocol tags: `post_p0`, `legacy_pre_review2` / `pre_p0_fix`, synthetic short-train.
4. Full-grid ~200-epoch JAMES Table claims: 待补充 (not invented).

2. Primary measured artefacts (post_p0)

Artefact	Role	SHA256 (first 16 hex)
<code>results/post_p0/ablation_summary.json</code>	Multi-seed mean±std for B1–M4, R0	c2ddc875a018a1ae
<code>results/post_p0/metrics_{ID}-s{0,1,2}.json</code>	Per-seed evaluation	(see §2.1)
<code>results/physics_ops/physics_ops_validation.json</code>	Stage E SQG / advection	c01f1054d6627029
<code>results/legacy_pre_review2/metrics_B2_GPU96.json</code>	Quarantined historical	4c9344624604903e

Protocol (ablation_summary): NATL60, crop_size=96, epochs=15, batch_size=1, seeds=[0,1,2], CUDA. Note in JSON: *not* full-grid JAMES Table rows. Git commit recorded in summary: 89f61b6.

2.1 Per-seed metric files

Present under `results/post_p0/`:

- B1-s0/s1/s2, B2-s0/s1/s2, M1-s0/s1/s2, M2-s0/s1/s2, M3-s0/s1/s2, M4-s0/s1/s2, R0-s0/s1/s2

Each JSON includes `ckpt`, `crop_size`, `seed`, `git_commit_ckpt`, `git_commit_eval`, `model.*`, `geostrophic.*`, and `protocol_note`.

2.2 Checkpoints (local only; not pushed)

Corresponding best weights: `checkpoints/4dvarnet-{ID}-s{seed}-best.pt` (excluded from git as *.pt). Training histories: `checkpoints/history-{ID}-s{seed}.json` (+ `-meta.json`).

3. How numbers were computed

Stage	Script	Inputs	Outputs
E	<code>scripts/validate_physics_operators.py</code>	NATL60 truth crop96	<code>results/physics_ops/*</code>
F–G train/eval	<code>scripts/run_post_p0_pipeline.py</code> → <code>scripts/train.py</code> , <code>scripts/evaluate.py</code>	<code>config/default.yaml</code> , <code>config/r0_compact_faithful.yaml</code>	ckpts + <code>results/post_p0/metrics_*.json</code>
F–G summarize	<code>pipeline</code> / <code>finalize_post_p0.py</code>	metrics JSONs	<code>ablation_summary.json</code>
H	<code>scripts/run_sensitivity.py</code>	B2-s0 / M3-s0 ckpts	<code>results/post_p0/sensitivity/</code> (if completed)
Figures	<code>scripts/plot_science.py</code> + <code>fourdvarnet/plotting.py</code>	metrics JSON	<code>results/figures/</code> , mirrored to <code>docs/paper/figures/</code> , <code>docs/report/figures/</code>

Stage	Script	Inputs	Outputs
Paper	scripts/assemble_paper.py	docs/paper/0*.md	PAPER.md, paper.html, paper.pdf
Report	scripts/build_report.py	metrics + PNGs	docs/report/report.{html,md,pdf}, root report.html

Core libraries: fourdvarnet/model.py, solver.py, physics.py, losses.py, metrics.py; **data** data/natl60.py, data/dataset.py.

4. Quarantine: legacy vs current

Tag	Path	Use
post_p0	results/post_p0/	Primary scientific ranking for this package
legacy_pre_review2 / pre_p0_fix	results/legacy_pre_review2/	Historical crop96/20ep only; do not cite as current claims
Synthetic short	results/metrics_B2.json etc.	Directional code-path only

Known historical issue (documented): pre-P0 geostrophic τ_{uv} could be pathological (~ -3.7); post_p0 OI-only geo $\tau_{uv} \approx \mathbf{0.846}$ (sane).

5. What was NOT run / 待补充

- Full-grid uncropped NATL60 ($\sim 200 \times 200$) $\times \sim 200$ epochs $\times$ multi-seed JAMES Table (4GB VRAM / wall-time).
- Byte-faithful CIA-Oceanix R0 + official pretrained checkpoint evaluation.
- VarDyn comparator experiments (separate codebase).
- Formal OSE / drifter score tables (hooks may exist; scores 待补充).
- Copying any external paper table as ours — **not done**.
- Stage H sensitivity: **Done** — results/post_p0/sensitivity/sensitivity_summary.json (SST factors 1/4/8; altimetry keep_every=3; strain bins). M3 τ_{uv} drops under strong SST coarsening (factor 8: ~ 0.876 vs ~ 0.909 at factor 1); sparsity impact mild.

6. Pytest / tooling gate

Command (faceswap env, 2026-09-15 package build):

```
python -m pytest -q --tb=line
..... [100%]
```

Status: **green** (58 tests).

7. Plagiarism / table hygiene

- Manuscript and report tables are built from **local** JSON only.
- Literature DOIs were verified via WebSearch (2026-09-15): Fablet 2024 10.1029/2023MS003609; Beauchamp 2023 10.5194/gmd-16-2119-2023; VarDyn 2025 10.1029/2024MS004689; Lapeyre & Klein 2006 10.1175/JPO2840.1.
- Paper body avoids local absolute paths and git SHAs (audit/report may include them).

8. Deliverable map

Deliverable	Paths
Outline	docs/paper/OUTLINE.md
Paper	docs/paper/PAPER.md, paper.html, paper.pdf
Report	docs/report/report.md, report.html, report.pdf (+ root report.html)
Audit	docs/AUDIT_EVIDENCE.md (+ optional html/pdf)

Deliverable	Paths
Figures	results/figures/, docs/paper/figures/, docs/report/figures/

9. Recompute hashes (maintainer)

```
python -c "import hashlib, pathlib; p=pathlib.Path('results/post_p0/ablation_summary.json'); print(hashlib.sha256(p.read_bytes()).hexdigest())"
```